

Rectangular matrix multiplication from polynomial products

Przemysław Uznański

Pathway

przemek@pathway.com

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Abstract

In this note, we extend the matrix-multiplication analysis in [9] to rectangular products and prove

$$\omega(1, k, 1) \leq \begin{cases} 2, & 0 \leq k \leq \frac{1}{2}, \\ 1 + k + \frac{1}{4k}, & k \geq \frac{1}{2}. \end{cases}$$

In particular, $\omega(1, \frac{1}{2}, 1) = 2$ and the dual exponent satisfies $\alpha \geq \frac{1}{2}$. The proof uses explicit degenerations of convolution tensors, multinomial counting, and Strassen’s spectral theorem. An angular constraint on three dot-product parameters gives both the rectangular curve and a bound for $\omega(a, b, c)$, with exact exponent $a + b$ whenever $a \geq b \geq c \geq 0$ and $c(a + b) \leq ab$. As an application, Zwick’s algorithm [15] for all-pairs shortest paths in directed unweighted graphs runs in $O(n^{2.5})$ time. Combining the rectangular bound with the $(\min, +)$ -product improvement of Alman and Vassilevska Williams [1] further gives $O(n^{2.4999})$ running time.

1 Introduction

Disclaimer. This note, its proofs, analysis and writeup were obtained fully through OpenAI Codex (GPT-6.1). The author has read and analyzed this note in full and takes full responsibility for its contents, and takes no credit for this work.

The initial motivation for this note was to explore whether the methods of the groundbreaking matrix-multiplication result in [9] could be pushed to analyze rectangular matrix products. We also aim to gather useful bounds and their algorithmic consequences in one place. Rephrasing the proofs in a form accessible to the theoretical computer science community has value in its own right.

Complexity convention. Throughout this note, all complexity bounds suppress a $+o(1)$ term in the exponent: $O(n^c)$ denotes $n^{c+o(1)}$. All shape parameters are fixed.

Rectangular matrix multiplication is a basic ingredient in algorithms for all-pairs problems, including shortest paths, lowest common ancestors, and Hamming distance. Its exponents depend on the shape of the product; an improvement for square matrices need not give a comparable improvement for every rectangular shape. The analysis in [9] establishes the square bound $\omega \leq 2.25$ using a shared-leg entropy inequality and degenerations of convolution tensors. We extend that analysis to rectangular products by retaining the orientation information lost under full leg symmetrization.

We work over $\mathbb{C}$. The resulting exponent bounds hold over every field of characteristic zero by invariance of asymptotic tensor rank under field extension [10, Theorem 3.10].

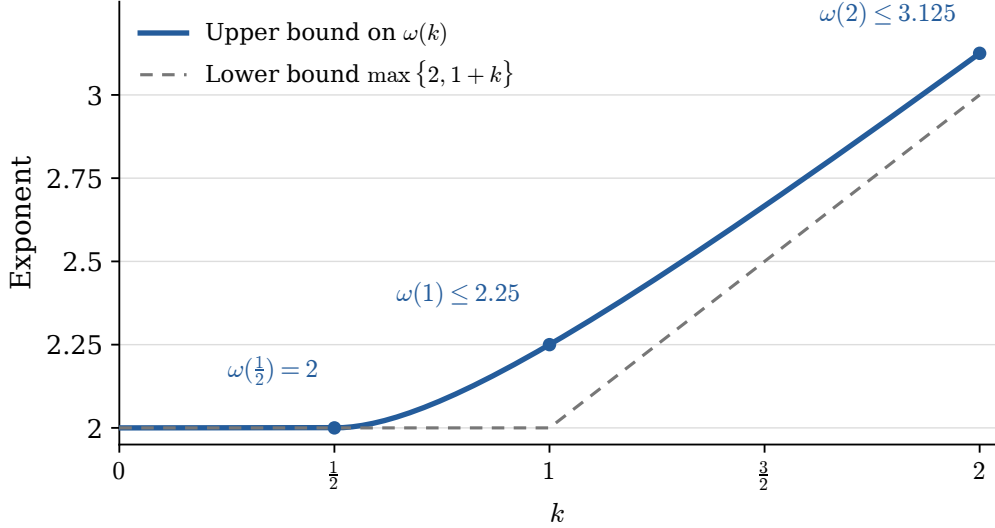


Figure 1: The upper bound on $\omega(k)$ from Theorem 1.1 for $0 \leq k \leq 2$, together with the size lower bound $\max\{2, 1+k\}$. The bounds coincide for $0 \leq k \leq \frac{1}{2}$.

For fixed nonnegative real u, v, w , let $\omega(u, v, w)$ be the infimum of the exponents e for which multiplying an $\lceil n^u \rceil \times \lceil n^v \rceil$ matrix by a $\lceil n^v \rceil \times \lceil n^w \rceil$ matrix uses $O(n^e)$ arithmetic operations, with the convention above. Write $\omega(k) = \omega(1, k, 1)$ and $\alpha = \sup\{k \geq 0 : \omega(k) = 2\}$.

Theorem 1.1. *For every fixed real $k \geq 0$,*

$$\omega(k) \leq \begin{cases} 2, & 0 \leq k \leq \frac{1}{2}, \\ 1 + k + \frac{1}{4k}, & k \geq \frac{1}{2}. \end{cases} \quad (1)$$

Consequently $\omega(k) = 2$ for $0 \leq k \leq \frac{1}{2}$, $\alpha \geq \frac{1}{2}$, and $\omega(2) \leq 3.125$.

Theorem 1.1 gives exponent two for inner dimension up to $n^{0.5}$, and an upper bound within $\frac{1}{4k}$ of the input-size lower bound $1+k$ for $k \geq 1$.

The proof has two parts. First, we list explicit ways to split and combine polynomial products and apply a tensor character to them. A character assigns numerical sizes that add on direct sums, multiply on tensor products, and cannot increase under restriction. Multinomial counting gives concavity and growth inequalities for these sizes. Rational rescaling, common slopes, and affine bounds then give $\sum_i \arccos \sqrt{p_i} \geq \pi/2$ for the character's three dot-product parameters.

Second, a sum-of-squares identity bounds weighted sums of these parameters. We apply Strassen's spectral theorem: the largest character value equals asymptotic tensor rank. We state precisely the form of this theorem we use, then show how the rank bound gives a running-time bound for every fixed matrix shape.

Relation to the preceding analysis. The two polynomial splittings in Lemma 2.1 recap the proofs of Lemmas 4.1 and 4.2 of [9] in coefficient coordinates. The finite separation adapts Proposition 3.1 of that reference, with a different root-of-unity filter using $2M$ copies instead of $5M$. All three constructions are proved in the appendices. Multinomial counting then gives the required power-sum inequality. The replication construction, rational rescaling, and coupling of six affine offsets supply the rectangular extension. The weighted bound uses the classical Wolstenholme inequality, for which we give a sum-of-squares proof. Strassen's

spectral theorem is a standard input; a modern exposition is [12]. We cite it rather than reproduce its proof. We make no priority claim for the standard tools or the finite construction variants. Section 7 substitutes the bounds into known algorithmic tradeoffs.

2 Polynomial products and finite constructions

A tensor is a trilinear form in three groups of variables, called its *legs*. A *restriction* makes linear substitutions independently on the three legs. A *degeneration* is a limit of such restrictions: the substitutions may depend on $\varepsilon \neq 0$, and every resulting coefficient converges as $\varepsilon \rightarrow 0$. Our constructions first change bases and then multiply variables by integer powers of ε . The resulting coefficient array is a polynomial in ε whose constant coefficient is the desired tensor. We write $T \geq S$ for a restriction and $T \supseteq S$ for a degeneration. Successive constructions can use a single parameter: replace the earlier parameter by a sufficiently large power of the later one, so every earlier error term still has positive total exponent.

The *rank* $\text{rank}(T)$ is the least number of products of three linear forms, one per leg, summing to T . The direct sum $S \oplus T$ uses disjoint variables on all three legs and represents independent computations. The tensor product $S \otimes T$ pairs their indices, as in a Kronecker product, and represents their composition on blocks. We write mT for m independent copies, and $T^{\otimes N}$ for N tensor factors. Thus $\text{rank}(T)xyz \geq T$. All factors used below restrict to xyz by keeping one nonzero term; this is what it means to drop a factor.

The oriented dot products are

$$B_X(q) = x \sum_{j=1}^q y_j z_j,$$

with $B_Y(q), B_Z(q)$ obtained by permuting the legs. They have rank q , and $B_i(q) \otimes B_i(q')$ is $B_i(qq')$ after reindexing. The matrix multiplication tensor factors as

$$\langle A, B, C \rangle = \sum_{u,v,w} x_{uv} y_{vw} z_{wu} \cong B_Y(A) \otimes B_Z(B) \otimes B_X(C). \quad (2)$$

Thus the three dot-product dimensions specify the matrix shape.

For distinct input legs i, j , with remaining output leg k , the *convolution tensor* is

$$C_{ij}(r, s) = \sum_{h=0}^{r-1} \sum_{\ell=0}^{s-1} x_h^{(i)} x_\ell^{(j)} x_{h+\ell}^{(k)}.$$

It multiplies two polynomials represented by their coefficient vectors. Its rank is $r + s - 1$: evaluation at $r + s - 1$ points and interpolation give the upper bound, and the independent output coefficients give the matching matrix-rank lower bound. Decreasing either input length is a restriction. Also

$$C_{ij}(1, s) \cong B_i(s), \quad C_{ij}(r, 1) \cong B_j(r). \quad (3)$$

Blocks T_h are *matched sectors sharing leg i* when $T = \sum_h T_h$ and the blocks may share variables on leg i , but use pairwise disjoint variables on each other leg. The following constructions hold for every permutation of the legs.

Lemma 2.1 (Finite separation and polynomial splittings). *1. If T has M matched sectors sharing leg i , then*

$$(2M)T \supseteq \bigoplus_{h=1}^M (T_h \otimes B_i(M)). \quad (4)$$

The sectors on the right are independent on all three legs.

2. For $r \geq 1$ and $s \geq 2$, $C_{ij}(r, s) \otimes B_i(2)$ degenerates to two matched sectors sharing leg i , namely $C_{ij}(r, s+1)$ and $C_{ij}(r, s-1)$.
3. For $r, s \geq 1$, $C_{ij}(r, 3s+r-1)$ degenerates to three matched sectors sharing leg i : two copies of $C_{ij}(r, s)$ and one copy of $C_{ik}(r, s)$.

Appendices A and B give the polynomial splittings in coefficient coordinates. Appendix C proves (4) using a finite geometric sum and a nonnegative square. The next construction splits a convolution into repeated independent copies of a smaller convolution.

Lemma 2.2 (Replicating a convolution). *For all positive integers m, r, s ,*

$$C_{ij}(mr, ms) \geq mC_{ij}(r, s). \quad (5)$$

The construction combines polynomial division with the standard evaluation description of cyclic convolution [5, Theorem 5 and Section 5.1]. We give the full proof in Appendix E.

3 Tensor characters and counting

3.1 Strassen's spectral theorem

A *tensor character* is a function λ from tensors to nonnegative real numbers satisfying

$$\begin{aligned} \lambda(S \oplus T) &= \lambda(S) + \lambda(T), & \lambda(S \otimes T) &= \lambda(S)\lambda(T), \\ \lambda(xyz) &= 1, & S \geq T &\implies \lambda(S) \geq \lambda(T). \end{aligned} \quad (6)$$

These rules express addition of independent computations, multiplication of tensor factors, and monotonicity under linear substitutions. Matrix rank after grouping one leg against the other two is an example. Keeping one nonzero term and using $\text{rank}(T)xyz \geq T$ gives $1 \leq \lambda(T) \leq \text{rank}(T)$ for every nonzero T .

Lemma 3.1 (Interpolation of a degeneration). *Suppose a degeneration $T \trianglerighteq S$ produces a tensor $P(\varepsilon)$ whose coefficients are polynomials of degree at most L . Then, for every integer $N \geq 1$,*

$$(NL+1)T^{\otimes N} \geq S^{\otimes N}. \quad (7)$$

Consequently, every tensor character satisfies $\lambda(S) \leq \lambda(T)$.

Equation (7) is the standard interpolation conversion from degeneration to restriction [2, Theorem 8], applied to tensor powers. We include the proof for completeness in Appendix D.

The *asymptotic rank* of T is

$$\tilde{R}(T) = \lim_{N \rightarrow \infty} \text{rank}(T^{\otimes N})^{1/N} = \inf_{N \geq 1} \text{rank}(T^{\otimes N})^{1/N}.$$

The limit exists because tensor rank is submultiplicative; apply the subadditive-sequence form of Fekete's lemma to its logarithm.

Theorem 3.2 (Strassen's spectral theorem). *For every nonzero tensor T over $\mathbb{C}$,*

$$\tilde{R}(T) = \max_{\lambda} \lambda(T), \quad (8)$$

where the maximum ranges over all tensor characters.

We use [10, Theorem 3.8]; see also [12, Corollary 3.30]. The easy direction follows from $\lambda(T)^N = \lambda(T^{\otimes N}) \leq \text{rank}(T^{\otimes N})$. The theorem supplies the opposite direction. Thus, to prove an upper bound on asymptotic rank, it suffices to bound every character value.

Fix an arbitrary character λ . Its *dot-product parameters* are $p_i = \log_2 \lambda(B_i(2)) \in [0, 1]$, for $i = X, Y, Z$. They determine the values at all integer dimensions:

$$\lambda(B_i(q)) = q^{p_i} \quad (q \geq 1). \quad (9)$$

To see this, put $m = \lfloor N \log_2 q \rfloor$. Monotonicity in dimension and $B_i(q)^{\otimes N} \cong B_i(q^N)$ give

$$2^{mp_i} \leq \lambda(B_i(q))^N \leq 2^{(m+1)p_i}.$$

Take N -th roots and let $N \rightarrow \infty$. The interval $[0, 1]$ follows from $1 \leq \lambda(B_i(2)) \leq 2$. In particular, (2) gives

$$\lambda(\langle A, B, C \rangle) = A^{p_Y} B^{p_Z} C^{p_X}. \quad (10)$$

Our aim is to constrain these three parameters for every character. Define its *convolution profiles* by

$$f_{ij}(r, s) = \lambda(C_{ij}(r, s)). \quad (11)$$

3.2 Words with prescribed multiplicities

Suppose T splits into s matched sectors $T_1, \dots, T_s$ sharing leg i . In $T^{\otimes N}$ a sector is specified by a word of N sector labels. Retain on the two nonshared legs the words with n_h occurrences of label h , where $\sum_h n_h = N$. There are

$$M = \binom{N}{n_1, \dots, n_s}$$

matched sectors. For example, multiplicities $(2, 1)$ give the words 112, 121, 211. Each sector is a product of the T_h with the prescribed multiplicities. Apply (4); now all legs distinguish the words, so their factors can be reordered independently. This gives

$$(2M)T^{\otimes N} \supseteq M \left(\bigotimes_h T_h^{\otimes n_h} \right) \otimes B_i(M). \quad (12)$$

Lemma 3.3 (Multinomial power-sum inequality). *If $T \supseteq \sum_{h=1}^s T_h$ is a splitting into matched sectors sharing leg i and $p_i > 0$, then*

$$\lambda(T)^{1/p_i} \geq \sum_h \lambda(T_h)^{1/p_i}. \quad (13)$$

For fully independent summands, $T \supseteq \bigoplus_h T_h$ gives $\lambda(T) \geq \sum_h \lambda(T_h)$ directly by additivity.

Proof. For every tuple $\sum_h n_h = N$, applying λ to (12) gives

$$2M\lambda(T)^N \geq M^{1+p_i} \prod_h \lambda(T_h)^{n_h}.$$

Put $a_h = \lambda(T_h)^{1/p_i}$. Cancel M and take the power $1/p_i$:

$$M \prod_h a_h^{n_h} \leq 2^{1/p_i} \lambda(T)^{N/p_i}.$$

There are at most $(N+1)^s$ multiplicity tuples. Summing and using the multinomial theorem gives

$$\left(\sum_h a_h \right)^N \leq (N+1)^s 2^{1/p_i} \lambda(T)^{N/p_i}.$$

For fixed s , this extra factor is polynomial in N . Taking N -th roots and letting $N \rightarrow \infty$ proves (13). $\square$

Rank bounds and restrictions give

$$1 \leq f_{ij}(r, s) \leq r + s - 1, \quad f_{ij}(r, s) = f_{ji}(s, r), \quad (14)$$

and coordinatewise monotonicity. The equality merely interchanges the two polynomial inputs of the same convolution.

Lemma 3.4 (Powered concavity and tripling). *For $p_i > 0$, the sequence $s \mapsto f_{ij}(r, s)^{1/p_i}$ is concave, and for all integers $r, s \geq 1$,*

$$f_{ij}(r, 3s + r - 1)^{1/p_i} \geq 2f_{ij}(r, s)^{1/p_i} + f_{ik}(r, s)^{1/p_i}, \quad (15)$$

where i, j, k are distinct. For $p_j > 0$, the sequence $r \mapsto f_{ij}(r, s)^{1/p_j}$ is also concave.

Proof. The two-sector splitting and Lemma 3.3 give, for $s \geq 2$,

$$2f_{ij}(r, s)^{1/p_i} = \lambda(C_{ij}(r, s) \otimes B_i(2))^{1/p_i} \geq f_{ij}(r, s+1)^{1/p_i} + f_{ij}(r, s-1)^{1/p_i}.$$

This is discrete concavity in s ; input interchange gives the other coordinate. The three-sector splitting and the same lemma give (15). $\square$

4 Rescaling the convolution profiles

Applying a character to Lemma 2.2 gives

$$f_{ij}(mr, ms) \geq mf_{ij}(r, s). \quad (16)$$

In particular, $f_{ij}(m, m) \geq m$, and monotonicity gives $f_{ij}(r, s) \geq \min\{r, s\}$. We need rescaled profiles only at positive rational lengths. Define

$$g_{ij}(r, s) = \sup_{\substack{q \in \mathbb{Q}_{>0} \\ qr, qs \in \mathbb{Z}_{>0}}} \frac{f_{ij}(qr, qs)}{q}. \quad (17)$$

Lemma 4.1 (Rescaled profiles). *The profiles g_{ij} are coordinatewise nondecreasing and satisfy*

$$\min\{r, s\} \leq g_{ij}(r, s) \leq r + s, \quad g_{ij}(r, s) = g_{ji}(s, r), \quad g_{ij}(tr, ts) = t g_{ij}(r, s). \quad (18)$$

For $p_i > 0$, g_{ij}^{1/p_i} is concave in s and satisfies

$$g_{ij}(r, 3s + r)^{1/p_i} \geq 2g_{ij}(r, s)^{1/p_i} + g_{ik}(r, s)^{1/p_i}. \quad (19)$$

For $p_j > 0$, g_{ij}^{1/p_j} is concave in r . Here all lengths, scale factors, and concavity coefficients are rational.

Proof. The bounds and input interchange follow from (14) and the lower bound above. Replacing q by q/t proves homogeneity.

The crucial point is that finitely many supremum values can be approached at one common scale. Choose scales q_h approximating the desired values, and choose Q so that every Q/q_h is a positive integer. By (16),

$$\frac{f_{ij}(Qr, Qs)}{Q} \geq \frac{f_{ij}(q_h r, q_h s)}{q_h}.$$

Replacing Q by an integer multiple preserves all these approximations and can clear any additional rational denominators. Monotonicity now follows by comparing lengths at a common scale.

For concavity, take $s = \tau s_1 + (1 - \tau)s_2$, with rational $0 \leq \tau \leq 1$. Approach $g_{ij}(r, s_1)$ and $g_{ij}(r, s_2)$ at one scale Q , with Qr, Qs_1, Qs_2, Qs integers. Discrete concavity gives

$$\left(\frac{f_{ij}(Qr, Qs)}{Q}\right)^{1/p_i} \geq \tau \left(\frac{f_{ij}(Qr, Qs_1)}{Q}\right)^{1/p_i} + (1 - \tau) \left(\frac{f_{ij}(Qr, Qs_2)}{Q}\right)^{1/p_i}.$$

Bound the left side by $g_{ij}(r, s)^{1/p_i}$ and let the approximation error tend to zero. Input interchange gives the other coordinate.

For tripling, monotonicity first enlarges the discrete source length $3s + r - 1$ to $3s + r$. At a common scale Q ,

$$\left(\frac{f_{ij}(Qr, Q(3s + r))}{Q}\right)^{1/p_i} \geq 2 \left(\frac{f_{ij}(Qr, Qs)}{Q}\right)^{1/p_i} + \left(\frac{f_{ik}(Qr, Qs)}{Q}\right)^{1/p_i}.$$

Approach the two right-hand suprema simultaneously; the left side is bounded by $g_{ij}(r, 3s + r)^{1/p_i}$. This proves (19). $\square$

5 Common slopes and the angular constraint

Index the three tensor legs by 1, 2, 3. We use the rescaled profiles to constrain the three dot-product parameters without averaging their orientations. Until (24), all profile arguments are positive rationals.

Lemma 5.1 (Common slopes and affine bounds). *Fix i with $p_i > 0$, and let j, k be the other legs. The functions $P_{ij}(s) = g_{ij}(1, s)^{1/p_i}$ and $P_{ik}(s) = g_{ik}(1, s)^{1/p_i}$ have a common asymptotic slope $L_i > 0$. Their normalized offsets*

$$c_{ij} = \sup_{s \in \mathbb{Q}_{>0}} \left(\frac{P_{ij}(s)}{L_i} - s \right), \quad c_{ik} = \sup_{s \in \mathbb{Q}_{>0}} \left(\frac{P_{ik}(s)}{L_i} - s \right)$$

are nonnegative and satisfy $c_{ij} + c_{ik} \leq 1$. Moreover,

$$P_{ij}(s + t) \geq P_{ij}(s) + L_i t, \quad P_{ij}(s) \leq L_i(s + c_{ij}),$$

and likewise for P_{ik} .

Proof. Write $P = P_{ij}$ and $Q = P_{ik}$. For $s > 1$, the nonnegative secant slope $(P(s) - P(1))/(s - 1)$ decreases as s grows, by concavity. It therefore has a finite limit, which is also $L_P = \lim P(s)/s$. Likewise define $L_Q = \lim Q(s)/s$. Dividing the coupled inequalities

$$P(3s + 1) \geq 2P(s) + Q(s), \quad Q(3s + 1) \geq 2Q(s) + P(s)$$

by s and passing to infinity gives $L_P \geq L_Q$ and $L_Q \geq L_P$. Call their common value L_i .

Their sum $R = P + Q$ satisfies $R(3s + 1) \geq 3R(s)$. Iterating at $s_m = 3^m(s + \frac{1}{2}) - \frac{1}{2}$ gives

$$R(s) \leq \lim_{m \rightarrow \infty} 3^{-m} R(s_m) = L_i(2s + 1). \quad (20)$$

Since $R(s) > 0$, we have $L_i > 0$. For $v > u > 0$, concavity bounds the secant slope $(P(v) - P(u))/(v - u)$ below by every later secant slope; sending its far endpoint to infinity gives

$$P(v) - P(u) \geq L_i(v - u).$$

Letting $u \downarrow 0$ through rationals and using $P(u) \geq 0$ gives $P(v) \geq L_i v$. Thus $P(s)/L_i - s$ is nonnegative and nondecreasing; the same holds for Q . Their sum is at most one by (20). Approaching both suprema at one sufficiently large rational s proves $c_{ij} + c_{ik} \leq 1$. The affine upper bounds follow from the definition of the offsets. $\square$

Proposition 5.2 (Angular constraint). *The dot-product parameters of every tensor character satisfy*

$$\sum_{i=1}^3 \arccos \sqrt{p_i} \geq \frac{\pi}{2}. \quad (21)$$

Proof. If some $p_i = 0$, its summand is already $\pi/2$. Assume all three parameters are positive. Lemma 5.1 gives six offsets and their three bounds

$$c_{ij} \geq 0, \quad c_{ij} + c_{ik} \leq 1. \quad (22)$$

The increment and affine bounds give

$$\frac{P_{ij}(s+t)}{P_{ij}(s)} \geq 1 + \frac{L_i t}{P_{ij}(s)} \geq 1 + \frac{t}{s + c_{ij}}.$$

Using homogeneity, we obtain

$$\frac{g_{ij}(r, s+t)}{g_{ij}(r, s)} \geq \left(1 + \frac{t}{s + c_{ij}r}\right)^{p_i}. \quad (23)$$

Increase r by εr , then s by εs . Input interchange supplies the first bound and (23) supplies the second:

$$\begin{aligned} 1 + \varepsilon &= \frac{g_{ij}((1+\varepsilon)r, (1+\varepsilon)s)}{g_{ij}(r, s)} \\ &\geq \left(1 + \frac{\varepsilon r}{r + c_{ji}s}\right)^{p_j} \left(1 + \frac{\varepsilon s}{s + c_{ij}(1+\varepsilon)r}\right)^{p_i}. \end{aligned}$$

Take logarithms, divide by ε , and let rational $\varepsilon \downarrow 0$. The elementary limit $\log(1 + \varepsilon x)/\varepsilon \rightarrow x$ gives

$$1 \geq \frac{p_i s}{s + c_{ij}r} + \frac{p_j r}{r + c_{ji}s} \quad (r, s \in \mathbb{Q}_{>0}). \quad (24)$$

With $t = r/s$, this is equivalent to

$$(1 - p_j)c_{ij}t^2 + (1 - p_i - p_j + c_{ij}c_{ji})t + (1 - p_i)c_{ji} \geq 0 \quad (t \in \mathbb{Q}_{>0}).$$

The polynomial is continuous, so the inequality holds for every real $t > 0$; no extension of the profiles to real lengths is needed. A quadratic $At^2 + Bt + C$ with $A, C \geq 0$ is nonnegative on $t > 0$ exactly when $B \geq -2\sqrt{AC}$, including zero coefficients. Thus writing $x = \sqrt{c_{ij}c_{ji}}$ gives $(x + \sqrt{(1 - p_i)(1 - p_j)})^2 \geq p_i p_j$, or

$$\sqrt{c_{ij}c_{ji}} \geq \left(\sqrt{p_i p_j} - \sqrt{(1 - p_i)(1 - p_j)}\right)_+, \quad (25)$$

where $(z)_+ = \max\{z, 0\}$.

For any nonnegative x_1, x_2, x_3 , AM–GM and (22) give

$$\sum_{i < j} 2\sqrt{c_{ij}c_{ji}}x_i x_j \leq \sum_{i < j} (c_{ij}x_i^2 + c_{ji}x_j^2) \leq \sum_i x_i^2. \quad (26)$$

Put $\theta_i = \arccos \sqrt{p_i}$ and suppose $S = \theta_1 + \theta_2 + \theta_3 < \pi/2$. Take $x_i = \cos \theta_i > 0$. By (25), $\sqrt{c_{ij}c_{ji}} \geq \cos(\theta_i + \theta_j)$. For the other two indices j, k , the addition formula gives

$$\cos(\theta_i + \theta_j)x_j + \cos(\theta_i + \theta_k)x_k = x_i + \cos S \cos(\theta_j - \theta_k) > x_i.$$

The last inequality uses $|\theta_j - \theta_k| \leq S < \pi/2$. Multiply by x_i and sum over i :

$$\sum_{i < j} 2\sqrt{c_{ij}c_{ji}}x_i x_j \geq \sum_{i < j} 2\cos(\theta_i + \theta_j)x_i x_j > \sum_i x_i^2,$$

contradicting (26). □

6 From character bounds to running times

Strassen's theorem turns the constraint on dot-product parameters into rank bounds. We first spell out how these bounds give algorithms.

Lemma 6.1 (Character bounds give algorithms). *Fix $\mathbf{h} = (h_X, h_Y, h_Z) \geq 0$. If the dot-product parameters of every tensor character satisfy $\sum_i h_i p_i \leq E$, then $\omega(h_Y, h_Z, h_X) \leq E$.*

Proof. The zero target is trivial. For a fixed integer $N \geq 2$, put $D_i = \lceil N^{h_i} \rceil$ and $M_N = \bigotimes_i B_i(D_i)$. Since $D_i \leq 2N^{h_i}$ and $0 \leq p_i \leq 1$, every character satisfies

$$\lambda(M_N) = \prod_i D_i^{p_i} \leq 8N^{\sum_i h_i p_i} \leq 8N^E.$$

Theorem 3.2 therefore gives

$$\tilde{R}(M_N) \leq 8N^E. \quad (27)$$

Fix any positive exponent slack η . First choose N so large that $8 < N^{\eta/2}$. The infimum defining $\tilde{R}(M_N)$ then gives a fixed integer K and an exact rank decomposition with

$$R := \text{rank}(M_N^{\otimes K}) \leq N^{K(E+\eta)}.$$

This is a finite bilinear algorithm for matrix dimensions (D_Y^K, D_Z^K, D_X^K) . Apply it recursively ℓ times to blocks. By (2), the resulting dimensions are $(D_Y^{K^\ell}, D_Z^{K^\ell}, D_X^{K^\ell})$ and there are R^ℓ scalar multiplications.

Each input and output dimension, viewed as a number of scalar entries, is at most R , by the corresponding matrix-rank lower bound. Thus the linear substitutions and additions over all recursion levels use $O(\ell R^\ell)$ operations. Given input size n , choose $\ell = \lceil \log_N n/K \rceil$ and pad to these dimensions. They cover the target shape because $D_i^{K^\ell} \geq n^{h_i}$, while $R^\ell = O(n^{E+\eta})$. This proves the bound for every positive slack, and hence the lemma under our exponent convention. All choices of N , K , and the finite identity precede the limit $n \rightarrow \infty$. $\square$

We now bound weighted sums of the dot-product parameters; the rectangular curve follows by substitution.

Corollary 6.2. *For every fixed $a \geq b \geq c \geq 0$,*

$$\omega(a, b, c) \leq \begin{cases} a + b, & c(a + b) \leq ab, \\ \frac{a + b + c}{2} + \frac{1}{4} \left(\frac{ab}{c} + \frac{ac}{b} + \frac{bc}{a} \right), & c(a + b) > ab. \end{cases} \quad (28)$$

When $c(a + b) \leq ab$, equality holds by the size lower bound.

Proof. Reorder the character's dot-product parameters so that the objective is $ap_1 + bp_2 + cp_3$, and write $p_i = \cos^2 \theta_i$ with $\theta_i \in [0, \pi/2]$. By Proposition 5.2, their sum is at least $\pi/2$. Decreasing angles to make the sum exactly $\pi/2$ only increases the objective, so it suffices to bound that case. If $b = 0$, then $c = 0$ and the objective is at most a .

For $a, b, c > 0$, put $\beta_i = 2\theta_i$ and

$$u = \sqrt{\frac{bc}{a}}, \quad v = \sqrt{\frac{ac}{b}}, \quad w = \sqrt{\frac{ab}{c}}.$$

Thus $vw = a$, $wu = b$, $uv = c$, and $\beta_1 + \beta_2 + \beta_3 = \pi$. The identity

$$\begin{aligned} u^2 + v^2 + w^2 - 2(a \cos \beta_1 + b \cos \beta_2 + c \cos \beta_3) &= (u - v \cos \beta_3 - w \cos \beta_2)^2 \\ &\quad + (v \sin \beta_3 - w \sin \beta_2)^2 \geq 0 \end{aligned} \quad (29)$$

follows by expansion and $\cos(\beta_2 + \beta_3) = -\cos \beta_1$. This proves the classical Wolstenholme inequality [8, Equation (1)], including zero angles. Since $p_i = (1 + \cos \beta_i)/2$, it gives

$$ap_1 + bp_2 + cp_3 \leq \frac{a+b+c}{2} + \frac{1}{4} \left(\frac{ab}{c} + \frac{ac}{b} + \frac{bc}{a} \right).$$

For $a, b > 0$, set $c_0 = ab/(a+b)$. If $0 \leq c \leq c_0$, replace c in the objective by c_0 , which only increases it. The displayed bound at c_0 equals $a+b$. This proves both branches, including $c = 0$.

We have bounded the objective for every tensor character. Lemma 6.1, with target $\mathbf{h} = (c, a, b)$, gives (28). $\square$

Proof of Theorem 1.1. Apply Corollary 6.2 to the sorted dimensions of $(1, k, 1)$. For $0 \leq k \leq 1$, the first branch applies exactly when $2k \leq 1$. In the remaining range, both possible orderings give

$$\frac{2+k}{2} + \frac{1}{4} \left(2k + \frac{1}{k} \right) = 1 + k + \frac{1}{4k}.$$

The output-size lower bound $\omega(k) \geq 2$ proves equality for $0 \leq k \leq \frac{1}{2}$ and hence $\alpha \geq \frac{1}{2}$. $\square$

Some values of the bound in Theorem 1.1 are:

k	0.5	0.6	0.8	1	1.5	2
$\omega(k) \leq$	2	2.0166...	2.1125	2.25	2.6666...	3.125

The exponent convention gives an $O(n^2)$ -time algorithm at $k = \frac{1}{2}$.

7 Applications

Theorem 1.1 and Corollary 6.2 give the following bounds through standard rectangular-multiplication balances. All bounds use arithmetic operations, with graph operations and symbol comparisons also charged.

1. **One lowest common ancestor per pair in a DAG.** Grandoni et al. [4, Theorem 4.2 of the published version] obtain exponent $1+2x$ from the balance $\omega(x) = 3x$. Theorem 1.1 permits $x = (1+\sqrt{3})/4$, giving $O(n^{(3+\sqrt{3})/2})$ time for finding one lowest common ancestor for every pair in an n -vertex DAG. The exponent is $(3 + \sqrt{3})/2 \approx 2.36603$.
2. **Exact all-pairs Hamming distance.** For n vectors of length n over an unrestricted alphabet, the reduction to dominance products [7] and Yuster's tradeoff [13, Proposition 2.1] give the balance $\rho = \omega(4 - \rho)$. Theorem 1.1 gives $\rho = (13 - \sqrt{7})/4 \approx 2.58856$ and running time $O(n^\rho)$ for computing every exact Hamming distance.
3. **The standard unweighted APSP balance.** Zwick's deterministic algorithm for directed unweighted APSP [15, Section 6] has exponent $2 + \mu$, where $\omega(\mu) \leq 1 + 2\mu$. Here $\omega(\frac{1}{2}) = 2 = 1 + 2(\frac{1}{2})$, so it gives $O(n^{2.5})$ running time. Combining the full curve with the deterministic $(\min, +)$ -product algorithm and APSP reduction of Alman and Vassilevska Williams [1, Theorems 22 and 33–34] further gives $O(n^{2.4999})$ running time.
4. **Minimum and maximum witnesses.** For two Boolean $n \times n$ matrices, a minimum (maximum) witness for entry (i, j) is the least (greatest) index k with $A_{ik} = B_{kj} = 1$. The standard blocking algorithm [3, Section 2.1] takes $O(n^{1-r+\omega(r)} + n^{2+r})$ time. Taking $r = \frac{1}{2}$ gives $O(n^{2.5})$ time for all witnesses.

5. **Sparse matrix multiplication.** For two $n \times n$ matrices with at most m nonzero entries each, the sparse–dense decomposition of Yuster and Zwick [14, Section 3] gives $O(m^2/n^r + n^{\omega(r)})$ time for $0 \leq r \leq 1$. Thus $m \leq n^{1.25}$ permits $O(n^2)$ time.
6. **Dynamic matrix inverse and reachability.** Substituting Corollary 6.2 into van den Brand–Nanongkai–Saranurak’s tradeoff [11, Theorem 4.2], with $u = 0.50323551$ and $v = 0.80667763$ in $\max\{u + v, \omega(1, u, v) - u, \omega(v) - v\}$, gives $O(n^{1.30992})$ worst-case time per entry update and query for maintaining the inverse of a nonsingular $n \times n$ matrix, after $O(n^{2.25})$ preprocessing. The associated Monte Carlo reduction gives the same bounds for directed reachability under edge updates, with $O(1)$ query time for a fixed pair [11, Corollary C.12 and Theorem C.16].

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A Splitting two polynomial products

We recap the two-sector construction used in [9, Lemma 4.1] as an explicit change of coordinates for pairs of polynomials.¹

Fix $r \geq 1$ and $s \geq 2$. The tensor $C_{XY}(r, s) \otimes B_X(2)$ computes

$$(f, (a, b)) \mapsto (fa, fb), \quad \deg f < r, \quad \deg a, \deg b < s.$$

Thus tensoring with $B_X(2)$ means multiplying two polynomials by the same f . We will change coordinates to split the second input into lengths $s+1$ and $s-1$.

For $e \geq 1$ and $\deg a, \deg b \leq e$, use

$$p = a + tb, \quad h = b - p_{e+1}t^e, \quad p_d = [t^d]p.$$

Here $[t^d]p$ denotes the coefficient of t^d . We have $\deg p \leq e+1$ and $\deg h \leq e-1$, and these coordinates are invertible: $b = h + p_{e+1}t^e$, $a = p - tb$. Use them on the input pair with $e = s-1$ and on the output pair with $e = r+s-2$. Multiplication by f becomes

$$(p, h) \mapsto (fp, fh + p_st^{s-1}(f - f_{r-1}t^{r-1})).$$

Scale input h by ε^{-1} and its output coordinate by ε . The map is now

$$(p, h) \mapsto (fp, fh + \varepsilon p_st^{s-1}(f - f_{r-1}t^{r-1})) \xrightarrow{\varepsilon \rightarrow 0} (fp, fh).$$

The independent polynomials p, h have lengths $s+1, s-1$, respectively. This gives $C_{XY}(r, s+1)$ and $C_{XY}(r, s-1)$ sharing the first leg, with disjoint second and third legs. Permuting the legs gives the other orientations.

B The three-sector splitting

We recap the construction from [9, Lemma 4.2]. In $C_{XY}(r, 3s+r-1)$, a term has $0 \leq x < r$ and output index $z = x+y$. Partition the second-input and output indices into three intervals:

	left	middle	right
Y	$[0, s-1]$	$[s, 2s+r-2]$	$[2s+r-1, 3s+r-2]$
Z	$[0, s+r-2]$	$[s+r-1, 2s+r-2]$	$[2s+r-1, 3s+2r-3]$

Scale middle Y variables by ε and middle Z variables by ε^{-1} , leaving the others unchanged. Since $z = y+x$ with $0 \leq x < r$, left and right Y indices stay in the corresponding Z intervals. Cross-sector terms thus connect middle Y to outer Z and acquire a factor ε ; matched terms remain unchanged. As $\varepsilon \rightarrow 0$, precisely three matched sectors sharing X remain.

Translations identify the outer sectors with $C_{XY}(r, s)$. The middle is

$$\sum_{u=0}^{r-1} \sum_{v=0}^{s-1} x_{r-1-u} y_{s+u+v} z_{s+r-1+v},$$

which is $C_{XZ}(r, s)$ after reversing its first-leg basis. This identifies the individual sector. In the fixed-word construction of (12), the sectors acquire independent first-leg variables before their individual bases are changed.

¹This is the degree-one case of the classical Clebsch–Gordan formula [6, Theorem 2.6.3, pp. 56–57 of the author version]. The coefficient calculation here does not require that formula.

C Finite separation of a shared leg

We adapt the construction of [9, Proposition 3.1] with an equal-sums filter, reducing the copy count from $5M$ to $2M$. It suffices to share leg X . Write

$$T = \sum_{h=1}^M \sum_{a,b,c} \mu_{habc} x_a y_{hb} z_{hc}.$$

Take $L = 2M$ copies and a primitive L -th root of unity ζ . In copy $\ell = 0, \dots, L-1$, make the substitutions

$$\begin{aligned} x_a^{(\ell)} &\mapsto \sum_{g=1}^M \zeta^{\ell g} X_{ag}, \\ y_{hb}^{(\ell)} &\mapsto \sum_{u=1}^M \zeta^{\ell(u-h)} Y_{hbu}, \\ z_{hc}^{(\ell)} &\mapsto \frac{1}{L} \sum_{v=1}^M \zeta^{-\ell v} Z_{hcv}. \end{aligned}$$

Summing the copies multiplies each coefficient by

$$\frac{1}{L} \sum_{\ell=0}^{L-1} \zeta^{\ell(g+u-h-v)}.$$

The geometric sum is one for exponent divisible by L and zero otherwise. Since $|g+u-h-v| \leq 2(M-1) < L$, it retains precisely $g+u=h+v$.

Now give X_{ag}, Y_{hbu}, Z_{hcv} respective weights $g^2, 2hu - h^2$, and $-2hv$. On every retained term their total is

$$g^2 + 2h(u-v) - h^2 = (g-h)^2 \geq 0.$$

The weight-zero terms have $g=h$ and $u=v$, so the limit is

$$\sum_{h,a,b,c,u} \mu_{habc} X_{ah} Y_{hbu} Z_{hcu} = \bigoplus_h (T_h \otimes B_X(M)).$$

Every leg now distinguishes h . Although individual weights may be negative, the transformed coefficients form a polynomial of degree at most $(M-1)^2$. This proves (4).

D Interpolation of a degeneration

For completeness, we give the standard interpolation proof [2, Theorem 8].

Proof of Lemma 3.1. Let $P(\varepsilon)$ be the tensor obtained from T by the degeneration's substitutions. Its coefficients are polynomials in ε of degree at most L , and $P(0) = S$. Thus L describes the chosen degeneration. For every $\varepsilon \neq 0$, $P(\varepsilon)$ is a restriction of T . Put $Q(\varepsilon) = P(\varepsilon)^{\otimes N}$. Each coefficient of Q is a product of N coefficients of P , so its degree is at most NL , and $Q(0) = S^{\otimes N}$.

Choose $NL+1$ distinct nonzero points $\varepsilon_0, \dots, \varepsilon_{NL}$. Lagrange interpolation, applied to every coefficient, gives

$$S^{\otimes N} = Q(0) = \sum_{j=0}^{NL} c_j Q(\varepsilon_j), \quad c_j = \prod_{\substack{0 \leq h \leq NL \\ h \neq j}} \frac{-\varepsilon_h}{\varepsilon_j - \varepsilon_h}.$$

Each $Q(\varepsilon_j)$ is a restriction of $T^{\otimes N}$. To form the displayed weighted sum from their direct sum, identify the corresponding variables on all three legs and scale the variables on one leg of copy j by c_j . These are allowed linear substitutions, proving (7). Applying a character and taking N -th roots gives

$$\lambda(S) \leq (NL + 1)^{1/N} \lambda(T).$$

Letting $N \rightarrow \infty$ proves $\lambda(S) \leq \lambda(T)$. No continuity assumption on λ is needed. $\square$

E Replicating a convolution

Proof of Lemma 2.2. Put $P(t) = t^m - 1$. Repeated polynomial division writes the inputs uniquely as

$$F = \sum_{a=0}^{r-1} F_a P^a, \quad G = \sum_{b=0}^{s-1} G_b P^b, \quad \deg F_a, \deg G_b < m.$$

These are invertible linear coordinates: the basis polynomials $t^d P^a$ have distinct leading degrees $ma + d$. On the output retain the analogous blocks H_c for $0 \leq c < r + s - 1$. They use $m(r + s - 1)$ independent coordinates, within the original output length $m(r + s) - 1$. Divide each $F_a G_b = Q_{ab} + P R_{ab}$ with $\deg Q_{ab}, \deg R_{ab} < m$. Scale input blocks a, b by $\varepsilon^{-a}, \varepsilon^{-b}$ and output block c by ε^c . The transformed product has

$$H_c = \sum_{a+b=c} Q_{ab} + \varepsilon \sum_{a+b=c-1} R_{ab}.$$

Let $\alpha_1, \dots, \alpha_m$ be the distinct roots of P . Since $Q_{ab}(\alpha_h) = F_a(\alpha_h)G_b(\alpha_h)$, the limit gives

$$H_c(\alpha_h) = \sum_{a+b=c} F_a(\alpha_h)G_b(\alpha_h) \quad (1 \leq h \leq m).$$

Evaluation at m distinct points is invertible on each block of degree less than m . The resulting coordinates therefore give m independent convolutions of lengths r, s . The evaluation step is the usual diagonalization of cyclic convolution [5, Theorem 5 and Section 5.1]; the displayed scalings prove the degeneration, with coefficient degree at most one. $\square$